

Jitendra Nath Naik

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RESEARCH PROFILE

2021 – Present

Research Scholar (Stochastic PDEs)

Indian Institute of Technology Goa | Supervisor: Dr. Lok Pati Tripathi

As a Research Scholar, my work addresses both the theoretical and computational challenges of Stochastic Partial Differential Equations (SPDEs).

- **Theoretical Analysis:** Investigating the well-posedness and regularity of solutions for SPDEs.
- **Numerical Methods:** Developing robust algorithms for SPDEs, with implementation and verification conducted via MATLAB and FreeFEM++.



PUBLICATIONS & PREPRINTS

2026

Existence, Uniqueness, and Pathwise Regularity for Multidimensional Semilinear SPDEs with Locally Lipschitz Coefficients and Rough Initial Data (Preprint)

Manuscript submitted for publication June 25, 2026

2026

Optimal Error Estimates of a Finite Element Method for Semilinear SPDEs with Additive Noise and Nonsmooth Initial Data (Preprint)

Manuscript submitted for publication

2026

Optimal Error Estimates of a Finite Element Method for Semilinear SPDEs with Multiplicative Noise and Rough Initial Data

Manuscript in preparation



EDUCATION

Current

Ph.D. in Mathematics

Indian Institute of Technology Goa, India

2016 – 2018

M.Sc. in Mathematics

Utkal University, Odisha, India | CGPA: 7.70

2013 – 2016

B.Sc. (Hons) in Mathematics

Utkal University, Odisha, India | Percentage: 75.25%

2013

Higher Secondary (12th)

Maharishi College of Natural Law, CHSE, Odisha | Percentage: 72.00%

2011

High School (10th)

Capital High School, BSE, Odisha | Percentage: 76.67%



QUALIFYING EXAMINATIONS & AWARDS

2020	CSIR - UGC NET <i>All India Rank: 95</i>
2020	GATE Mathematics <i>All India Rank: 173</i>
2016	P.G. Meritorious Scholarship <i>Institute of Mathematics and Applications (IMA), Government of Odisha</i> Awarded a state-level merit scholarship for postgraduate studies in mathematics.



TECHNICAL STRENGTHS

Programming	C, C++, Python
Software	MATLAB, FreeFEM++, LaTeX



WORKSHOPS & FDPS

Jul 2024	Workshop on Num. Algorithms & Computations in Science & Tech. <i>Organized by IIT Ropar</i>
Jun 2024	FDP on Financial Math, SPDE Theory, and Numerical Trends <i>Organized by VIT-AP University (Online Mode)</i>
Feb 2023	High-End Workshop on Stochastic Modeling Technique <i>Organized by IIT(ISM) Dhanbad</i>
Nov 2022	Instructional School for Teachers on Partial Differential Equation <i>Organized by Tezpur University</i>
Sep 2022	Numerical Methods for Partial Differential Equations <i>Organized by National Center of Mathematics and IISER Thiruvananthapuram</i>
Jul 2022	High-End Workshop on Computational Finance <i>Organized by IIT Goa</i>



TEACHING ASSISTANTSHIP (IIT GOA)

Spring 24, 25	Ordinary Differential Equations (MA103)
Autumn 2024	Differential Equations II (MA302)
2022, 2024	Calculus (MTH101)
2023	Stochastic Processes
2022	Complex Analysis (MTH3151)
2022	Measure and Probability

2022

Numerical Analysis



EXTRA-CURRICULAR ACTIVITIES

2024

2nd Position in Cricket

Chakravyuh Sports Tournament

Held by the Sports Council at IIT Goa.